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THÁNG SINH VIÊN NGHIÊN CỨU KHOA HỌC CÔNG NGHỆ & SÁNG TẠO

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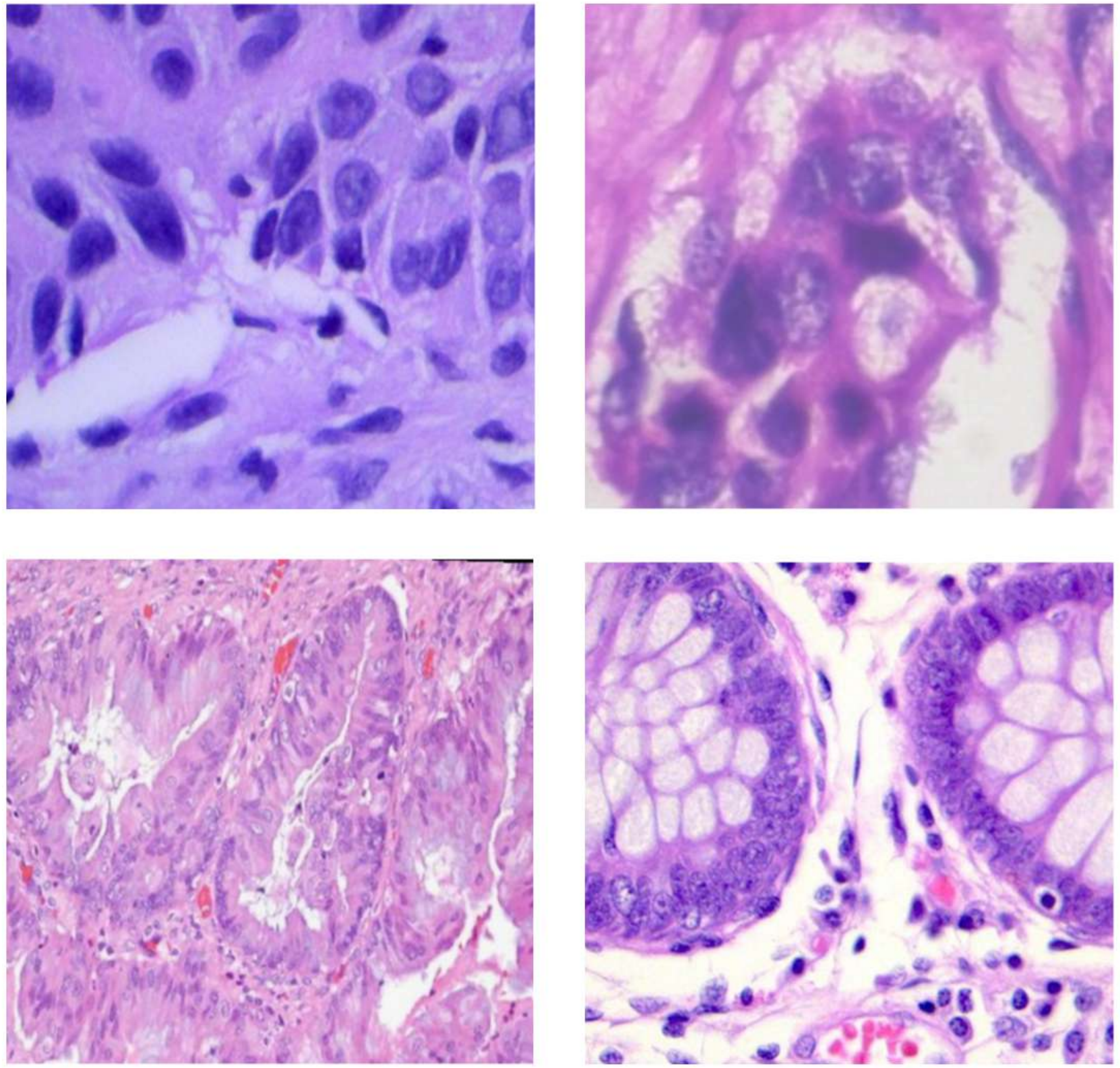
CODAPath: Contrastive Optimization via Dual-Foundation Models for Active Learning in Pathology

Nam-Hai Tran, Vu-Trung-Kien Nguyen, Tien-Dung Pham, Minh-Hai Bui,
Thi-Oanh Nguyen
School of Information and Communications Technology,
Hanoi University of Science and Technology

Problem



- High-capacity pathology models are powerful but **data-hungry**.
- Annotation by certified pathologists is **expensive**.

Active Learning (AL) offers a promising solution by selecting the most informative images for annotation

Related Work

Traditional AL

Evaluates uncertainty and coverage via learned model weights

Without pre-existing weights, forced to rely on random sampling

Modern AL

Bypasses naive initialization by data distribution and density

Focuses only on either uncertainty or coverage

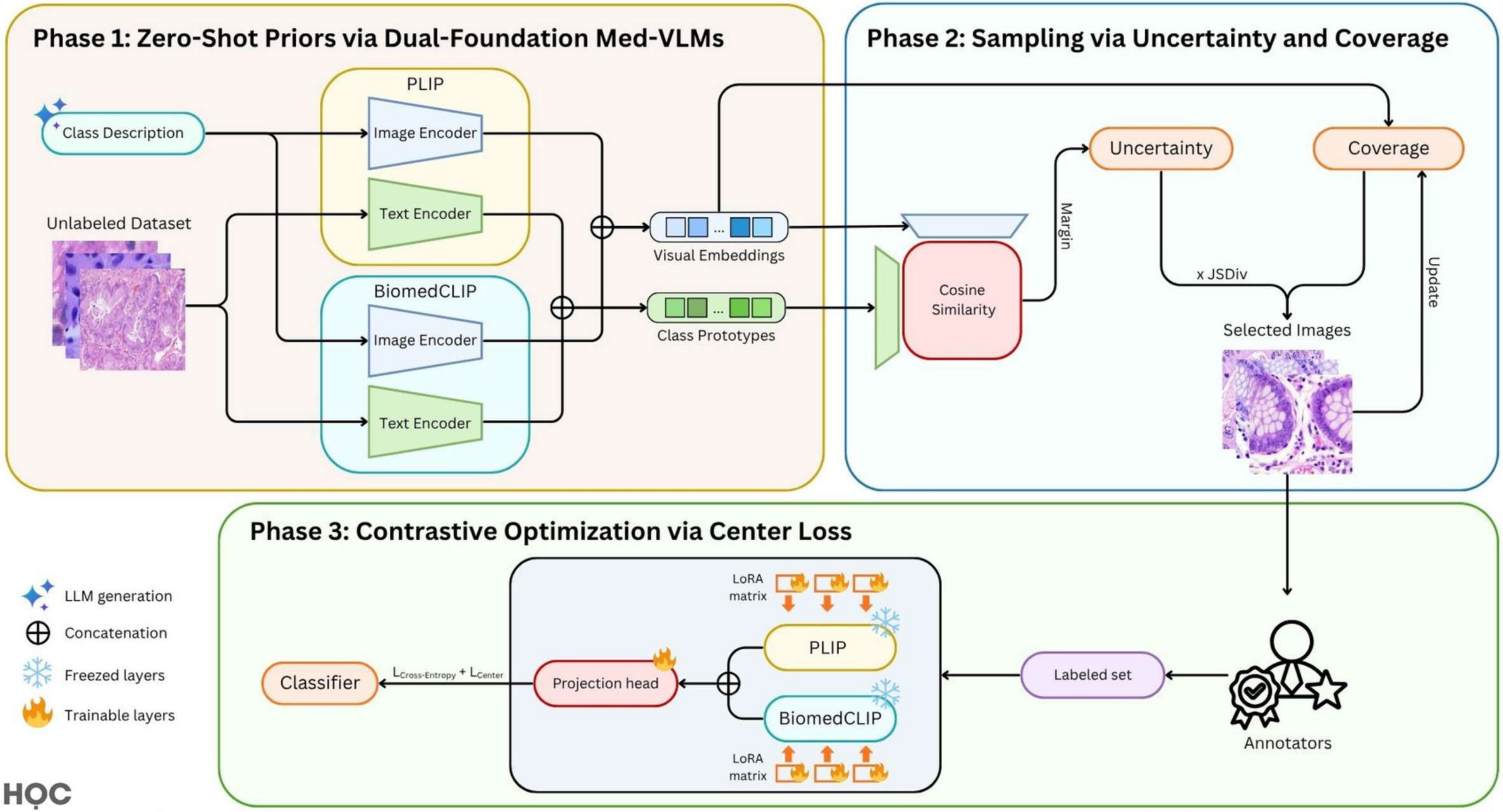
Both fails under **extreme low budget constraints**

CODAPath: A New Paradigm

Our key contributions

1	2	3
Using Dual Med-VLMs' prior knowledge to overcome random initialization	Defining a Scoring Metric to quantify both Uncertainty and Coverage for Sample Selection	Integrating Center Loss to tackle representation collapse under extreme data scarcity

CODAPath Framework



CODAPath Framework

- Phase 1: Zero-Shot Priors via Dual-Foundation Med-VLMs
- Phase 2: Sampling via Uncertainty and Coverage
- Phase 3: Contrastive Optimization via Center Loss

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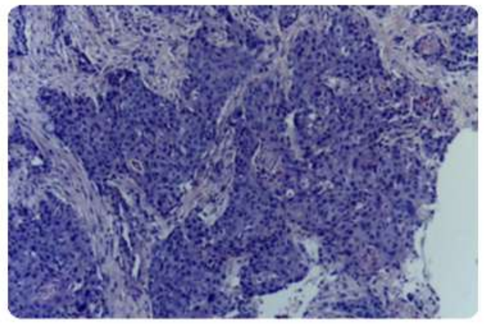
Phase 1: Zero-Shot Priors via Dual-Foundation Med-VLMs



oral_oscc

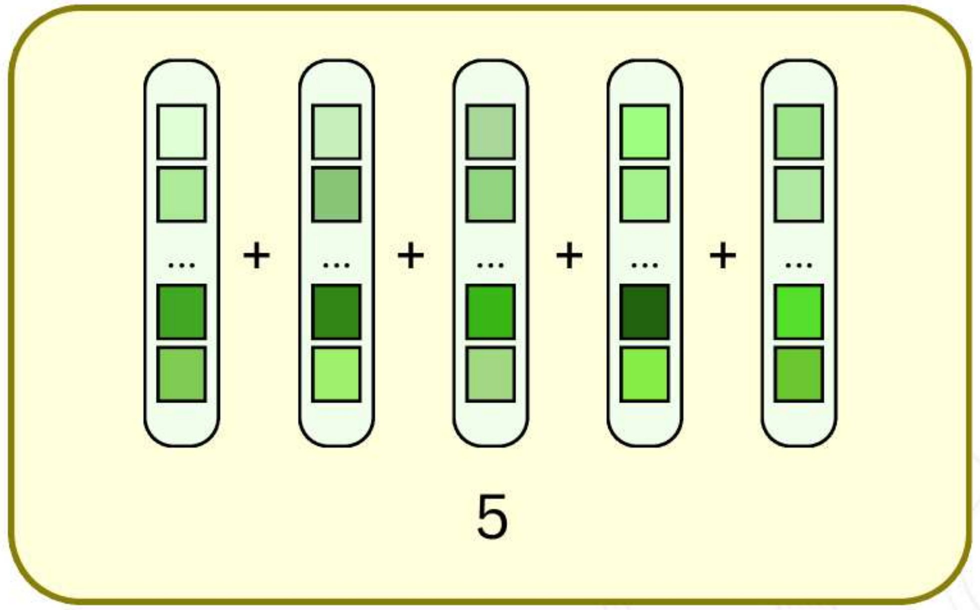
Gemini

***CLS_DESC**: oral squamous cell carcinoma, showing invasive malignant squamous cells with keratin pearls.*

Prompt templates

- Photomicrograph showing *[CLS_DESC]*
- Histopathology of *[CLS_DESC]*
- H&E stained microscopic image of *[CLS_DESC]*
- Pathological tissue section demonstrating *[CLS_DESC]*
- Cellular morphology characteristic of *[CLS_DESC]*

Text Encoder
PLIP ⊕ BiomedCLIP



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Class Prototype

Phase 1: Zero-Shot Priors via Dual-Foundation Med-VLMs

To obtain a strong representation that is
both detailed and comprehensive:

PLIP[1]
Highly **specialized** in pathology morphology from medical social media data.

BiomedCLIP[2]
Robust across **broad** biomedical contexts and high-res medical scans.

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Phase 2: Sampling via Uncertainty and Coverage

Deriving zero-shot class predictions from the image-text cosine similarity scores:

$$P_{i,c} = \frac{\exp(\hat{\mathbf{v}}_i^\top \hat{\mathbf{t}}_c / \tau)}{\sum_{j=1}^{C} \exp(\hat{\mathbf{v}}_i^\top \hat{\mathbf{t}}_j / \tau)}$$

- $\hat{\mathbf{v}}_i$: Visual embeddings of sample i; $\hat{\mathbf{t}}_c$: Class prototype for class c

Evaluating the model's confusion between the highest $P_{i,(1)}$ and second-highest $P_{i,(2)}$ probabilities:

$$m_i = P_{i,(1)} - P_{i,(2)}$$

Phase 2: Sampling via Uncertainty and Coverage

Penalizing uninformative prediction across all classes using Jensen-Shannon Divergence J_i:

$$J_i = \frac{1}{2}\sum_{c=1}^{C}\left[P_{i,c}\ln\left(\frac{P_{i,c}}{M_{i,c}}\right) + Q_{i,c}\ln\left(\frac{Q_{i,c}}{M_{i,c}}\right)\right]$$

- $Q_{i,c}$: Represents one-hot hard-label projection
- $M_{i,c} = 0.5(P_{i,c} + Q_{i,c})$

Finding the samples the model is most “confused” about:

$$U_i = (1 - m_i)(1 + J_i)$$

Phase 2: Sampling via Uncertainty and Coverage

Measures how much "new" spatial territory the candidate covers that is not already represented by the existing selected set

$$G(\mathbf{x}_k|S) = \sum_{i=1}^{|\mathcal{U}|} \max\left[0, s_{i,k} - \max_{x_j \in S} s_{i,j}\right]$$

- S : The set of currently selected images
- $\mathcal{U}$: The set of unlabeled pool
- $s_{i,j}$: The similarity score between sample x_i and x_j

Phase 2: Sampling via Uncertainty and Coverage

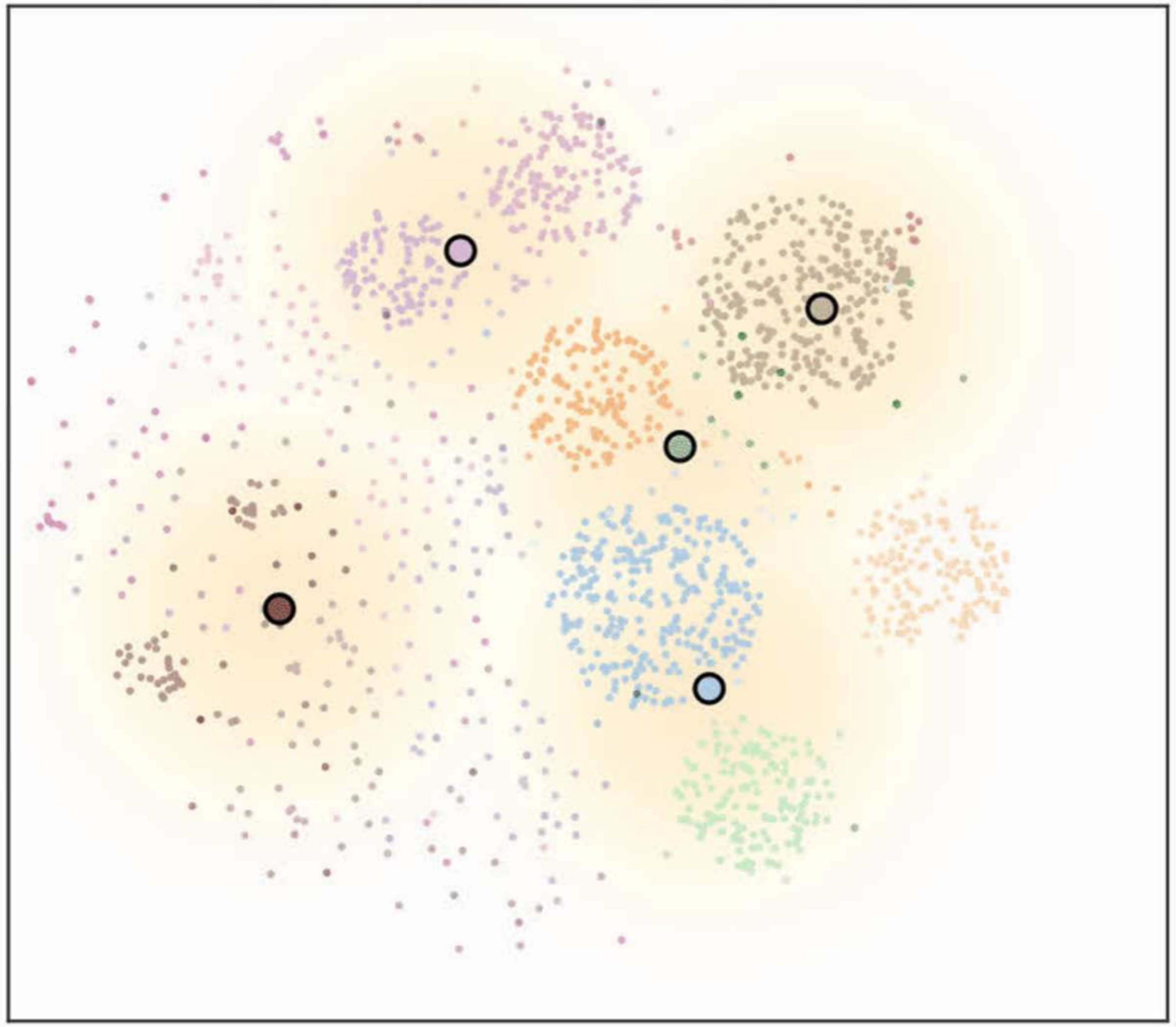
Select the sample $\mathbf{x}^*$ with the highest score in weighted average function:

$$\mathbf{x}^* = \underset{\mathbf{x}_k \in \mathcal{U} \setminus S}{\arg\max} \left[(1-\alpha) \cdot \tilde{G}(\mathbf{x}_k | S) + \alpha \cdot \tilde{U}_k \right]$$

Phase 2: Sampling via Uncertainty and Coverage

Require: Unlabeled pool $\mathscr{U}$, budget B.

Initialize selected set $S \leftarrow \emptyset$.
while $|S| < B$ **do**
    for each candidate $\mathbf{x}_k \in \mathscr{U} \setminus S$ **do**
        Compute uncertainty U_i
        Compute spatial coverage gain $G(\mathbf{x}_k|S)$
        Calculate weighted score
    end for
    Select $\mathbf{x}^*$ that has the highest score.
    $S \leftarrow S \cup \{\mathbf{x}^*\}$.
end while



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Phase 3: Contrastive Optimization via Center Loss

To train effectively with scarce data (e.g., 50 images), we use **LoRA** combined with **Center Loss**:

$$L = L_{CE} + \lambda L_{center}$$

$$L_{center} = \frac{1}{2N} \sum_{i=1}^{N} ||\hat{\mathbf{f}}_{proj,i} - c_{y_i}||_2^2$$

Center Loss pulls features of the same class towards their class center, increasing intra-class compactness and widening the margin between different classes.

Experimental Setup: Datasets

PathMNIST[3]

- Domain: 9 distinct colorectal tissue types
- Size: 107,180 images

SkinTissue[4]

- Domain: 16 diverse dermatopathology classes including tumors and normal structures
- Size: 129,364 images

HistoSet-5x14[5]

- Domain: 14 classes across multiple organ systems requiring subtle benign/malignant distinction
- Size: 28,000 images

Experimental Setup: Baselines

- Random selection
- Uncertainty-based methods: Margin[6], Entropy[7]
- Coverage-based methods: CoreSet[8]
- Hybrid methods balancing uncertainty and coverage: BADGE[9]
- Modern cold-start methods rely on unsupervised distribution matching: TypiClust[10], ActiveFT[11]

Comparison with existing methods

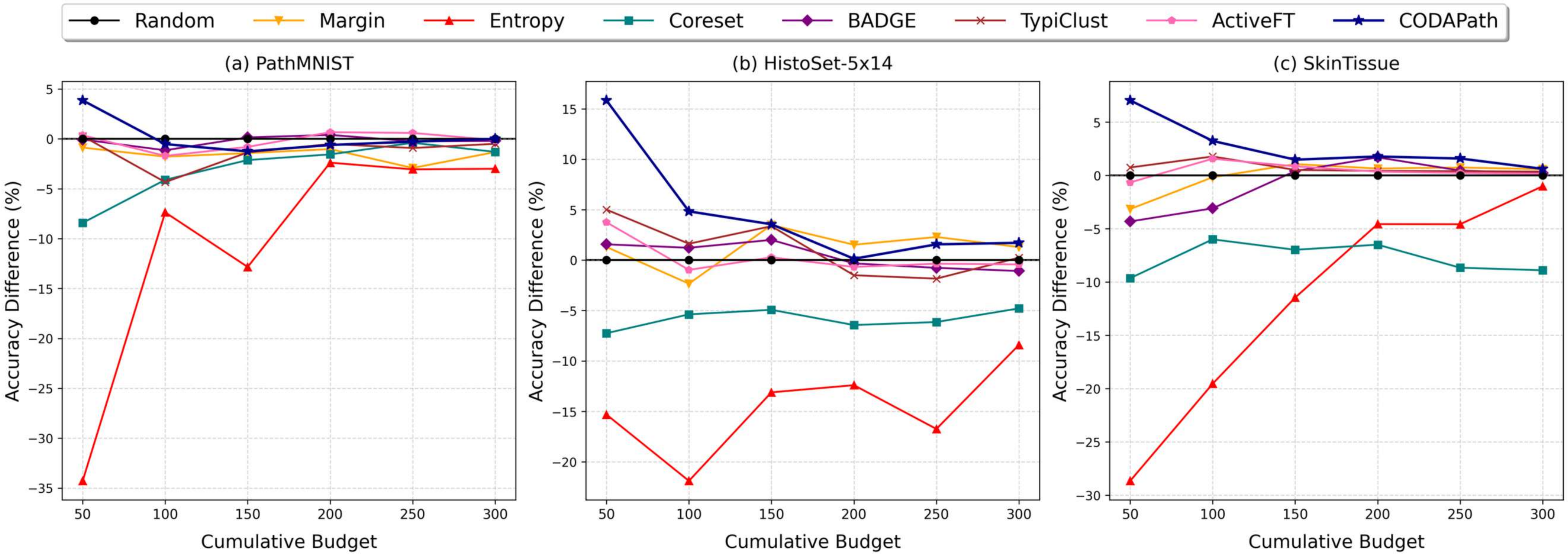
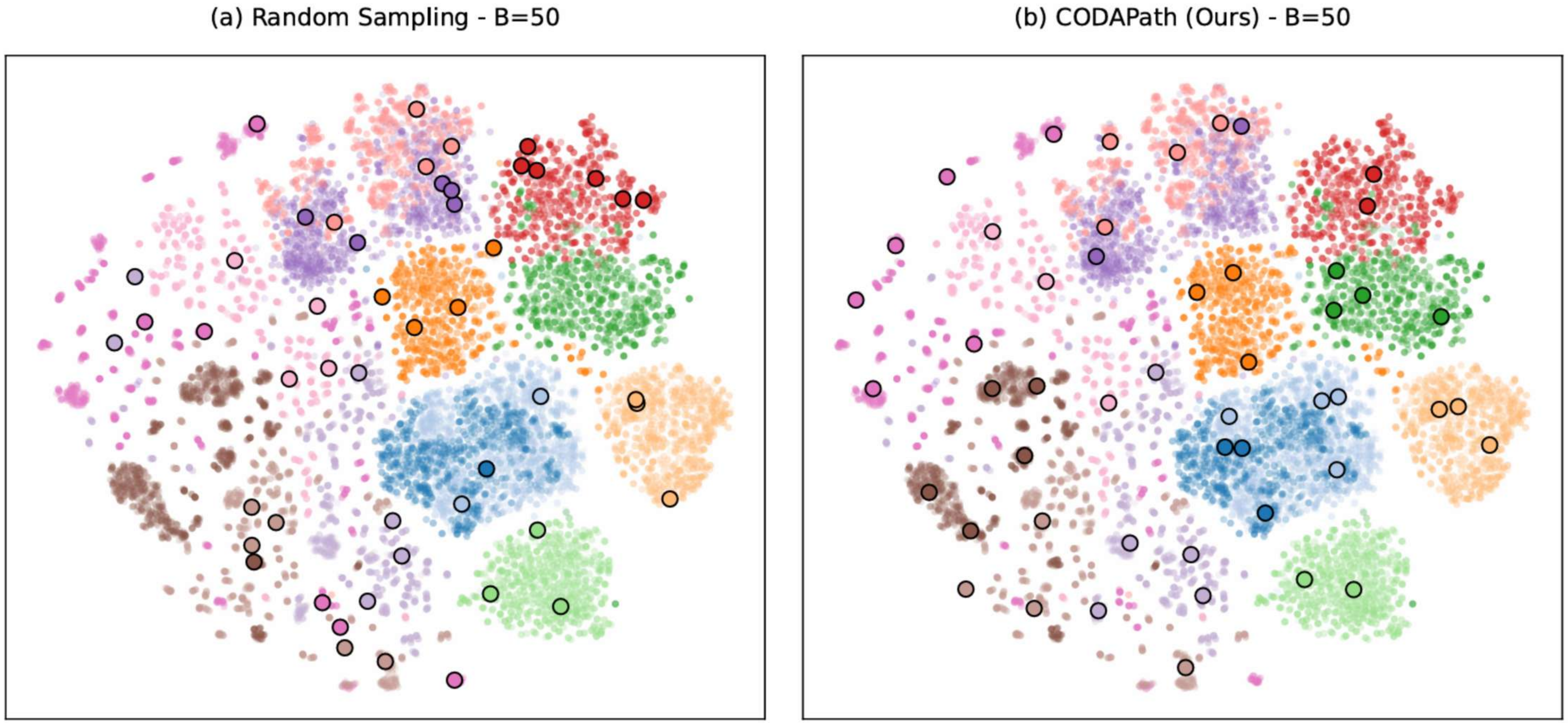


*Fig 1. Mean accuracy difference between several AL algorithms and random sampling across different budgets. A **positive** mean difference indicates that the corresponding AL method is **beneficial**.*

Sampling Visualization (t-SNE)



Random: Clustered together, missing sparse data regions.

CODAPath: Evenly covers the entire space, selecting diverse samples.

Ablation Study: Phase 1

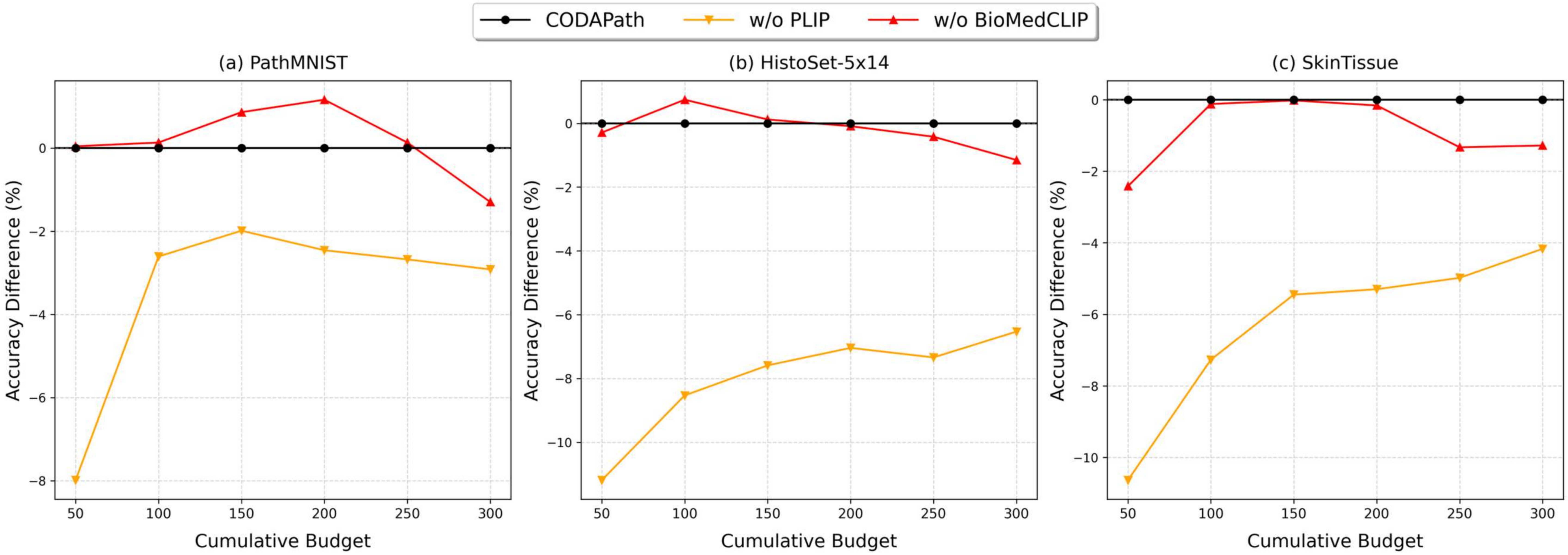


*Fig 2. Ablation study on Phase 1. The figure illustrates the accuracy **degradation** when either PLIP or BiomedCLIP is removed from the initialization process.*

Ablation Study: Phase 2

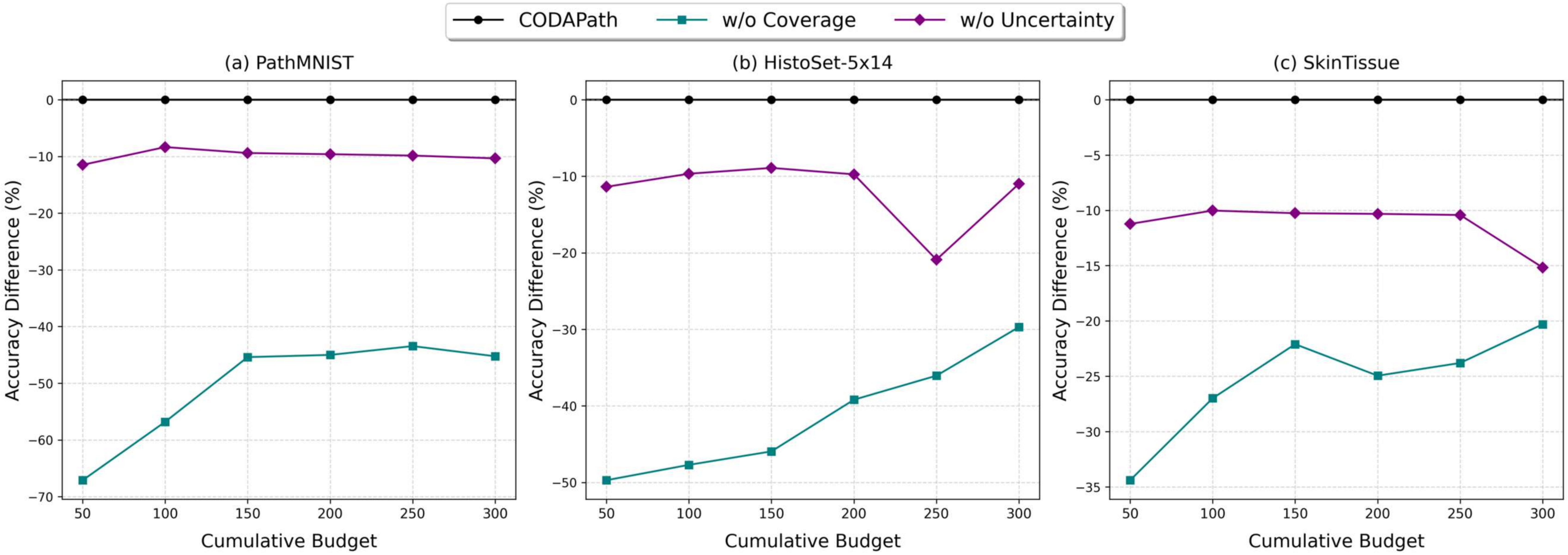


Fig 3. Ablation study on Phase 2. The figure illustrates the accuracy ***degradation*** *when the acquisition strategy relies solely on uncertainty (w/o coverage) or solely on coverage (w/o uncertainty).*

Ablation Study: Phase 3

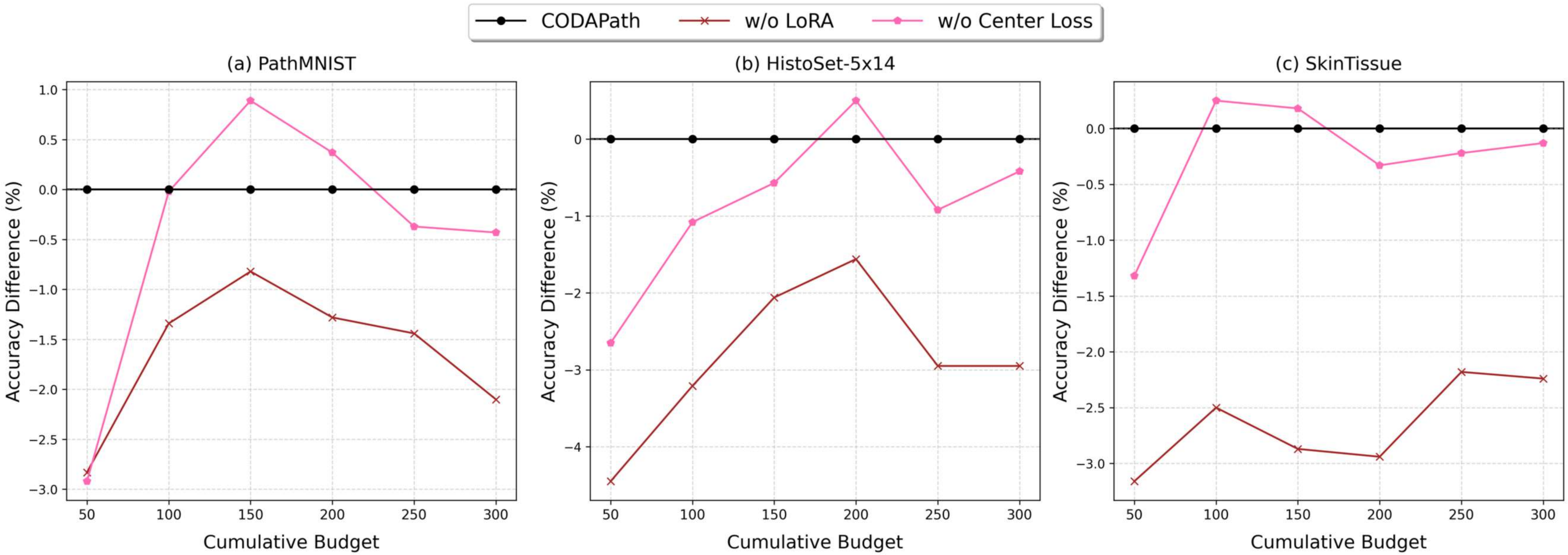


*Fig 5. Ablation study on Phase 3. The figure illustrates the accuracy **degradation** when removing LoRA or Center Loss during the final optimization phase.*

Conclusion

- Breakthrough: Reduce random initialization in medical Active Learning.
- Efficiency: Leverages Zero-shot knowledge from VLMs and detailed clinical descriptions.
- Robustness: Enables robust learning even with an extremely low annotation budget via Center Loss.

CODAPath establishes a powerful baseline for Active Learning in pathology.

Thank you!

Please do not hesitate to ask any questions

Our code:

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